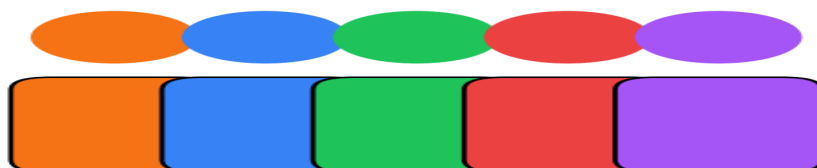


Investigating Fairness in Our Community



Task Information

Subject	Mathematics — MYP Year 1
Unit Title	Unit 2: Data With Reasoning
Key Concept / Related Concept	Logic / Representation & Justification
Global Context	Fairness and Development
GC Exploration	Inequalities, differences and inclusion
Inquiry Statement	How can showing data clearly help us explain patterns and changes in our communities?
ATL Skills Focus	Media Literacy Skills & Thinking Skills
Objectives Assessed	Criterion C: Communicating Criterion D: Applying Mathematics in Real-Life Contexts
Task Type	Project-Based Summative Assessment (Individual) — Data Investigation with A3 Infographic Chart & Oral Presentation
Duration	1 week (independent data collection time)



Scenario

Communities are not always fair — some neighbourhoods have more parks than others, some students have more access to the internet than others, some sports teams get more playing time than others. As a Year 1 mathematician and data researcher, your job is to investigate ONE real fairness or inclusion issue in your own school or community, collect real data about it, and use statistics and clear visual representation to explain what is really happening — and to help others understand it too.

You will present your findings as an A3 infographic chart and a short presentation to the class, so that your data can genuinely help explain patterns and changes to the people around you.



Part A – Plan & Collect (Criterion D focus)

- 1 Choose a fairness / inclusion issue in your school or local community to investigate through data (e.g. access to green spaces, internet access at home, participation in sports by gender, availability of recycling bins, access to books/library resources, playground time, canteen food choices, access to transport).
- 2 Write ONE clear inquiry question your data will help answer (link it to the unit inquiry statement).
- 3 State your population and describe the sample you will actually survey/observe (who, how many, why this sample is fair and unbiased).
- 4 Design a simple survey with 4–6 closed-ended questions OR plan a structured observation sheet to collect your data.
- 5 Collect your own primary data from at least 20 respondents/observations.



Green spaces



Internet access



Sports



Recycling



Library books



Transport



Canteen food



Part B – Analyse Your Data (Criterion C & D)

- 1 Organise your raw data into a frequency table with clear categories.
- 2 Classify your data as discrete or continuous and justify your classification.
- 3 Calculate the mean, median, mode and range of your data set (show all working).
- 4 Explain what the range tells us about spread/dispersion in your data, and connect this to the idea of 'fairness' or 'inequality' in your context.
- 5 Draw at least THREE different graphical representations of your data by hand or digitally (choose from: bar chart, pictogram, histogram, line plot, pie chart). Choose the graphs that best represent your data type.
- 6 Add ONE simple probability connection: identify a chance event linked to your topic and describe its likelihood using probability words (impossible, unlikely, even chance, likely, certain) or by conducting a small chance experiment (dice/coin/spinner) and comparing predicted vs actual results.



Part C – Build Your A3 Infographic Chart (Criterion C focus)

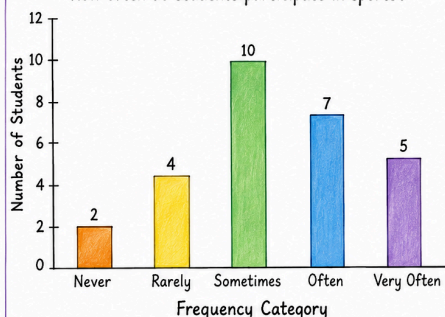
- 1 Design ONE A3-sized infographic poster that presents your investigation clearly and persuasively.
- 2 Your A3 chart MUST include: a title/inquiry question, your sample & population description, a frequency table, at least 2 graphs (correctly labelled with titles, axes and units), your mean/median/mode/range clearly displayed, and a short 'What this tells us about fairness in our community' conclusion box.
- 3 Use icons, colour-coding and labels effectively – think like a media designer, not just a mathematician (ATL: Media Literacy).
- 4 Reference your data source(s) and be honest about any limitations of your sample (e.g. small sample size, bias).
- 5 Example graph

DATA: How often do students participate in sports?

FREQUENCY CATEGORY	Never	Rarely	Sometimes	Often	Very Often
NUMBER OF STUDENTS	2	4	10	7	5

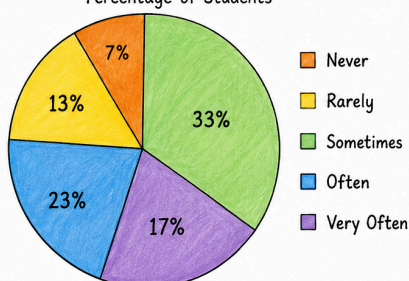
GRAPH 1: BAR CHART

How often do students participate in sports?



GRAPH 2: PIE CHART

Percentage of Students



This data shows how often 30 students participate in sports.



Part D – Present to the Class (Criterion C & D)

- 1 Present your A3 chart to the class in a 3–4 minute presentation (individual).
- 2 Explain: your inquiry question, how you collected the data, what your graphs and statistics show, and how your findings connect to fairness/inequality in your community.
- 3 Be ready to answer 2 questions from the teacher or classmates about your data and reasoning.



A3 Chart Requirements – Self-Check Before Submission

☐	Title / inquiry question clearly stated
☐	Population & sample described (who, how many, why chosen)
☐	Survey/observation method briefly explained
☐	Frequency table included
☐	At least 2–3 correctly labelled graphs (title, axes, units, key/legend)
☐	Mean, median, mode and range shown with working
☐	Range/dispersion explained in words
☐	One probability connection included (likelihood words or dice/coin/spinner experiment: predicted vs actual)
☐	Icons, colour-coding and clear labels used (media literacy)
☐	'What this tells us about fairness' conclusion box
☐	Data source & limitations acknowledged
☐	Neat, organised, and readable from 2 metres away



Suggested Timeline

Day 1	Launch task. Choose community fairness issues. Draft inquiry question, define population & sample.
Day 2	Design survey/observation sheet (closed-ended questions). Teacher checks before data collection begins.
Day 3	Collect data from at least 20 respondents/observations.
Day 4	Build frequency table. Classify discrete vs continuous. Calculate mean, median, mode, range.
Day 5	Draw graphs (choose best-fit types). Run a probability mini-experiment (dice/coin/spinner).

IB MYP 1
Summative Assessment - 2 (Sept 2026)

Day 6	Design and build A3 infographic chart. Peer feedback using the checklist.
Day 7	SA Presentations (3-4 minutes each) + Q&A. Submit A3 chart.



Assessment Rubric – Criterion C: Communicating

Level	Level Descriptor
0	The student does not reach a standard described by any of the descriptors below.
1-2	- Uses limited or no mathematical language on the A3 chart or in the presentation. - Uses limited forms of representation (e.g. only one graph, or no frequency table) to organise data, with limited success.
3-4	- Uses some appropriate mathematical language (e.g. mean, median, mode, range, discrete/continuous) with some accuracy. - Uses adequate forms of representation (frequency table + at least one graph) to organise data, with some success. - Communicates a line of reasoning that is difficult to follow in places.
5-6	- Uses appropriate mathematical language accurately across the poster and presentation. - Uses different, appropriate forms of representation (frequency table, 2-3 correctly labelled graphs, statistics) to organise data effectively. - Communicates a clear line of reasoning through the A3 chart and oral presentation. - Organises data using tables, graphs and diagrams that are mostly logical and coherent.
7-8	- Consistently uses appropriate and precise mathematical language, both in writing on the A3 chart and when speaking to the class. - Uses different, appropriate and well-chosen forms of representation (frequency table, multiple correctly labelled graphs, icons and infographic elements) to organise data clearly and persuasively. - Communicates a complete, coherent and concise line of reasoning that convincingly links data to the fairness/inequality context. - Organises information logically using tables, graphs and diagrams that follow a conventional, media-literate infographic format.



Assessment Rubric – Criterion D: Applying Mathematics in Real-Life Contexts

Level	Level Descriptor
0	The student does not reach a standard described by any of the descriptors below.
1-2	- Identifies elements of the real-life community issue with limited relevance. - Applies mathematical strategies (mean/median/mode/range, graphs) to the real-life situation with limited success, achieving little accuracy.
3-4	- Identifies some relevant elements of the community fairness issue chosen. - Selects and applies mathematical strategies to model the situation, reaching a solution that is partially correct. - Describes whether the solution makes sense in the context of the community issue, with limited justification.
5-6	- Identifies the relevant elements of the authentic community issue and constructs a survey/observation appropriate to it. - Selects and applies correct mathematical strategies (frequency tables, mean/median/mode/range, appropriate graph types, basic probability) to reach a valid, mostly accurate solution. - Explains the degree of accuracy of results and whether they make sense, referencing sample size or possible bias.
7-8	- Clearly identifies and thoughtfully explains the relevant elements of an authentic, real-life fairness/inequality issue in their own community. - Consistently selects and correctly applies a range of mathematical strategies (frequency tables, mean/median/mode, range as a measure of dispersion, correctly chosen graph types, probability language/experiment) to reach a correct, well-justified solution. - Justifies the degree of accuracy of the data and conclusions, critically discussing limitations (sample vs population, bias, sample size) and explains clearly whether the solution makes sense in real life. - Draws an insightful, well-justified conclusion connecting the data findings to fairness, inclusion or inequality in the community.



ATL Skills – Media Literacy & Thinking

Media Literacy: designing an infographic that communicates data clearly using icons, colour, layout and labels for a real audience.

Thinking Skills: interpreting patterns in data, drawing justified conclusions, and evaluating the reliability/fairness of a sample.



Student Checklist – Before You Submit

☐	I created at least two different representations.
☐	My graphs have titles, labels, scales and units.
☐	I calculated mean, median, mode and range for both groups.
☐	I checked my calculations.
☐	I explained what my statistics mean rather than only listing them.
☐	I compared Group A and Group B using mathematical evidence.
☐	My fairness conclusion uses at least three pieces of evidence.
☐	My recommendation is connected to the data.
☐	I used correct mathematical vocabulary and notation.
☐	I can explain my work in a 2-minute Presentation.

You've got this – good luck, data detective! 